

运动主机WIFI操作SOP

一、运动主机wifi模组说明

机器狗开机后，wifi驱动会加载初始化两个接口p2p0、wlan0，其中p2p0作为AP模式会向外发射wifi信号，即搜到的机器狗的wifi，默认密码为12345678。电脑连接机器狗wifi即可通过ssh进入机器狗的Ubuntu操作系统。wifi默认为36信道的5G频段。wlan0接口作为客户端模式，用于运动主机连接外部的wifi（比如外部局域网或公网）。

二、机器狗wifi名称或频段修改

非技术人员请勿随便操作，否则wifi配置文件一旦出错，可能需要通过重刷系统镜像解决

1、电脑连接机器狗wifi

2、ssh进入运动主机

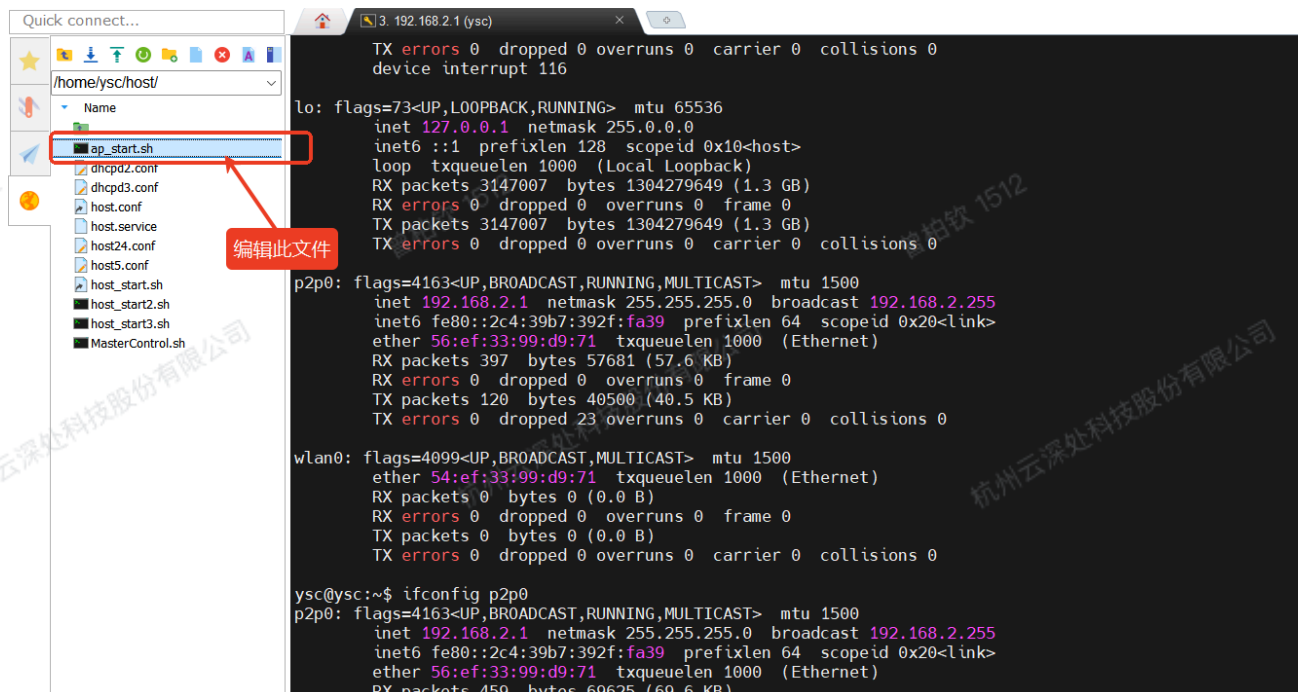
3、进入host目录



复制代码

```
cd host
```

4、编辑ap_start.sh文件来修改wifi或密码



```

ap_start.sh
1 #!/bin/bash
2
3 # con-name
4 CC_NAME_24G=myap24G
5 CC_NAME_5G=myap50G
6
7 dhcpip=192.168.2.1/24
8 SSID_PSK=12345678
9 # 5G channel 36 40 44 48 52~64 100~104 149~165
10 channel=36
11 apdevice=""
12
13 if [ -d /sys/class/net/wlan0 ];then
14     apdevice=wlan0
15 fi
16
17 if [ -d /sys/class/net/p2p0 ];then
18     apdevice=p2p0
19 fi
20
21 if [ $apdevice == "" ];then
22     echo "no wifi device"
23     exit 0
24 fi
25
26 id=$(cat /etc/boardinfo/sn)
27 SSID_50G=lite3_1385_5g
28 SSID_24G=yztest
29
30 function create_ap_5G()
31 {
32     nmcli radio wifi on
33     iw reg set CN
34     nmcli c add type wifi ifname $apdevice con-name $CC_NAME_5G autoconnect no ssid $SSID_50G
35     nmcli c modify $CC_NAME_5G 802-11-wireless.mode ap 802-11-wireless.band a 802-11-wireless.channel $channel
36     nmcli c modify $CC_NAME_5G 802-11-wireless-security.key-mgmt wpa-psk
37     nmcli c modify $CC_NAME_5G 802-11-wireless-security.psk $SSID_PSK
38     nmcli con modify $CC_NAME_5G ipv4.addresses $dhcpip
39     nmcli connection up $CC_NAME_5G
40 }

```

5、编辑host_start2.sh来修改wifi频段

Quick connect...

/home/ysc/host/

Name

..

ap_start.sh

dhcpcd2.conf

dhcpcd3.conf

host.conf

host.service

host24.conf

host5.conf

host_start.sh

host_start2.sh

host_start3.sh

MasterControl.sh

3. 192.168.2.1 (ysc)

TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

device interrupt 116

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536

inet 127.0.0.1 netmask 255.0.0.0

inet6 ::1 prefixlen 128 scopeid 0x10<host>

loop txqueuelen 1000 (Local Loopback)

RX packets 3147007 bytes 1304279649 (1.3 GB)

RX errors 0 dropped 0 overruns 0 frame 0

TX packets 3147007 bytes 1304279649 (1.3 GB)

TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

p2p0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500

inet 192.168.2.1 netmask 255.255.255.0 broadcast 192.168.2.255

inet6 fe80::2c4:39b7:392f:fa39 prefixlen 64 scopeid 0x20<link>

ether 56:ef:33:99:d9:71 txqueuelen 1000 (Ethernet)

RX packets 397 bytes 57681 (57.6 KB)

RX errors 0 dropped 0 overruns 0 frame 0

TX packets 120 bytes 40500 (40.5 KB)

TX errors 0 dropped 23 overruns 0 carrier 0 collisions 0

wlan0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500

14

15 iptables -t nat -A POSTROUTING -s 192.168.2.0/24 -o eth1 -j MASQUERADE

16 iptables -t nat -A POSTROUTING -s 192.168.137.0/24 -o wlan0 -j MASQUERADE

17 bash /home/ysc/host/ap_start.sh start 5G

18

当前为默认5G，若改为2.4G，则将5G改为24G

注意!!! 修改的话，是改为24G（不要带点）

6、进行第4步或者第五步的修改后，需要整机重启生效

三、机器狗连接外部wifi

1、执行iw dev确认wlan0接口可用且空闲

```

ether 56:ef:33:99:d9:71 txqueuelen 1000 (Ethernet)
RX packets 397 bytes 57681 (57.6 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 120 bytes 40500 (40.5 KB)
TX errors 0 dropped 23 overruns 0 carrier 0 collisions 0

wlan0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
ether 54:ef:33:99:d9:71 txqueuelen 1000 (Ethernet)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ysc@ysc:~$ ifconfig p2p0
p2p0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 192.168.2.1 netmask 255.255.255.0 broadcast 192.168.2.255
inet6 fe80::2c4:39b7:392f:fa39 prefixlen 64 scopeid 0x20<link>
ether 56:ef:33:99:d9:71 txqueuelen 1000 (Ethernet)
RX packets 459 bytes 69625 (69.6 KB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 143 bytes 45462 (45.4 KB)
TX errors 0 dropped 23 overruns 0 carrier 0 collisions 0

ysc@ysc:~$ cd host
ysc@ysc:~/host$ iw dev
phy#0
    Interface p2p0
        ifindex 10
        wdev 0x2
        addr 56:ef:33:99:d9:71
        ssid lite3_1385_5g
        type AP
        txpower 22.00 dBm
    Interface wlan0
        ifindex 9
        wdev 0x1
        addr 54:ef:33:99:d9:71
        type managed
        txpower 42949572.96 dBm
ysc@ysc:~/host$

```

执行

2、搜索附近wifi

复制代码

```
sudo nmcli device wifi list --rescan yes
```

```

Interface p2p0
    ifindex 10
    wdev 0x2
    addr 56:ef:33:99:d9:71
    ssid lite3_1385_5g
    type AP
    txpower 22.00 dBm
Interface wlan0
    ifindex 9
    wdev 0x1
    addr 54:ef:33:99:d9:71
    type managed
    txpower 42949572.96 dBm
ysc@ysc:~/host$ sudo nmcli device wifi list
[sudo] password for ysc:
IN-USE BSSID SSID MODE CHAN RATE SIGNAL BARS SECURITY
* 56:EF:33:99:D9:71 lite3_1385_5g Infra 36 0 Mbit/s 0 WPA1 WPA2

IN-USE BSSID SSID MODE CHAN RATE SIGNAL BARS SECURITY
* E6:63:17:DB:87:BE robotdog-5G Infra 1 270 Mbit/s 79 *** WPA2
18:FD:74:B9:A3:B9 ysc-m20-test Infra 40 270 Mbit/s 70 *** WPA1 WPA2
5C:A7:21:72:B6:93 wifi@dr.cn Infra 11 130 Mbit/s 65 *** WPA1 WPA2
40:FE:95:D2:B5:63 IT Infra 149 270 Mbit/s 65 *** WPA1 WPA2
5C:A7:21:72:B6:90 yscs Infra 11 130 Mbit/s 64 *** WPA1 WPA2
5C:A7:21:72:B6:96 Menjin Infra 11 130 Mbit/s 64 *** WPA1 WPA2
40:FE:95:D2:B5:66 Menjin Infra 149 270 Mbit/s 64 *** WPA1 WPA2
40:FE:95:D2:B5:64 sdwanceshi Infra 149 270 Mbit/s 64 *** WPA1 WPA2
40:FE:95:D2:B5:65 wifi@dr.cn Infra 149 270 Mbit/s 64 *** WPA1 WPA2
40:FE:95:D2:B5:61 guest@dr.cn Infra 149 270 Mbit/s 64 *** --
40:FE:95:D2:B5:60 wifi@dr.cn 5G Infra 149 270 Mbit/s 62 *** WPA1 WPA2
8A:49:2D:0E:1F:34 CR1B1_N0.17_5G Infra 36 54 Mbit/s 60 *** WPA2
40:FE:95:D2:B5:62 yscs Infra 149 270 Mbit/s 60 *** WPA1 WPA2
3C:67:C8:74:FA:61 BTWIFI_FA61 Infra 9 135 Mbit/s 55 ** WPA2
FC:A0:5A:1A:8B:AA OrayBox-2.4G-8BA8 Infra 6 130 Mbit/s 54 ** WPA2
8A:49:2D:0E:1E:EA CR1B1_N0.14_5G Infra 36 54 Mbit/s 50 ** WPA2
5A:14:21:80:18:EA AP00000329 Infra 1 65 Mbit/s 49 ** WPA2
5C:A7:21:72:C7:56 Menjin Infra 1 130 Mbit/s 47 ** WPA1 WPA2
A8:B5:8E:80:F6:A2 CA9B_N0.215_5G Infra 40 65 Mbit/s 47 ** WPA1 WPA2
5C:A7:21:72:C7:53 wifi@dr.cn Infra 1 130 Mbit/s 45 ** WPA1 WPA2
5C:A7:21:72:C7:50 yscs Infra 1 130 Mbit/s 45 ** WPA1 WPA2

```

准备连接这个

3、连接外部wifi，双引号内的名称或密码替换实际的即可



复制代码

```
sudo nmcli device wifi connect "名称" password "密码" ifname wlan0
```

4、

```
ysc@ysc:~/host$ sudo nmcli device wifi connect "robotdog-5G" password " " ifname wlan0
Device 'wlan0' successfully activated with '13cc99b4-d1a4-47b9-b84a-3014f85da9f5'.
ysc@ysc:~/host$ iw dev
phy#0
    Interface p2p0
        ifindex 10
        wdev 0x2
        addr 56:ef:33:99:d9:71
        ssid lite3_1385_5g
        type AP
        channel 1 (2412 MHz), width: 20 MHz, center1: 2412 MHz
        txpower 21.00 dBm
    Interface wlan0
        ifindex 9
        wdev 0x1
        addr 54:ef:33:99:d9:71
        ssid robotdog-5G
        type managed
        channel 1 (2412 MHz), width: 20 MHz, center1: 2412 MHz
        txpower 25.00 dBm
ysc@ysc:~/host$ ifconfig wlan0
wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.102.89 netmask 255.255.255.0 broadcast 192.168.102.255
    inet6 240e:471:8a0:883:d17c:1b08:7ba9:76a3 prefixlen 64 scopeid 0x0<global>
    inet6 240e:471:8a0:883:8368:c542:30de:8654 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::3829:84da:55b:f9f2 prefixlen 64 scopeid 0x20<link>
    ether 54:ef:33:99:d9:71 txqueuelen 1000 (Ethernet)
    RX packets 154 bytes 31375 (31.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1161412 bytes 340044247 (340.0 MB)
    TX errors 0 dropped 1 overruns 0 carrier 0 collisions 0
```

已连接

5、如果连接的是公网，可确认是否有网络



复制代码

```
ping www.baidu.com -4 -c 3
#-4代表指定使用ipv4请求百度网站，-c 3只ping3行
```

```
ysc@ysc:~/host$ ping www.baidu.com -4 -c 3
PING www.a.shifen.com (180.101.49.44) 56(84) bytes of data:
64 bytes from 180.101.49.44 (180.101.49.44): icmp_seq=1 ttl=51 time=59.1 ms
64 bytes from 180.101.49.44 (180.101.49.44): icmp_seq=2 ttl=51 time=48.4 ms
64 bytes from 180.101.49.44 (180.101.49.44): icmp_seq=3 ttl=51 time=41.9 ms
--- www.a.shifen.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 41.896/49.810/59.107/7.094 ms
ysc@ysc:~/host$
ysc@ysc:~/host$
```

6、附部分nmcli指令

查看网络连接状态	<code>sudo nmcli connection show</code>
断开wifi连接	按ssid断开: <code>sudo nmcli connection down "wifi名称的"</code> 按网卡断开: <code>sudo nmcli dev disconnect wlan0</code>
彻底删除网络（避免开机自连）	<code>sudo nmcli connection delete "wifi名称"</code>